

Problem Set 5

Solutions by Harsh Bhalala

Problem 1. Fibonacci Divisibility

(a)

Proof by strong induction. Let $P(n) := F_{n+a} \equiv F_n \cdot F_{a+1} \pmod{F_a}$, now to prove $p(n)$ for $n \geq 0$ by strong induction.

Base Case:

$$P(0) = F_a \equiv F_0 \cdot F_{a+1} \equiv 0 \pmod{F_a}.$$

$$P(1) = F_{a+1} \equiv F_1 \cdot F_{a+1} \equiv 1 \cdot F_{a+1} \equiv F_{a+1} \pmod{F_a}$$

Therefore, both cases hold.

Inductive step: For strong induction on n assume $P(n-1)$ and $P(n-2)$ holds for $n \geq 2$. So,

$$F_{(n-1)+a} \equiv F_{n-1} \cdot F_{a+1} \pmod{F_a} \quad (1)$$

$$F_{(n-2)+a} \equiv F_{n-2} \cdot F_{a+1} \pmod{F_a} \quad (2)$$

Now, by the properties of the congruence on equation (1) and (2),

$$F_{(n-1)+a} + F_{(n-2)+a} \equiv F_{n-1} \cdot F_{a+1} + F_{n-2} \cdot F_{a+1} \pmod{F_a}$$

$$F_{n+a} \equiv (F_{n-1} + F_{n-2}) \cdot F_{a+1} \pmod{F_a}$$

$$F_{n+a} \equiv F_n \cdot F_{a+1} \pmod{F_a}$$

Which implies $P(n)$ is true. Therefore, by the strong induction hypothesis, $P(n)$ is true for all $n \geq 0$. $\square$

(b)

Proof by induction. Induction hypothesis Assume $P(n)$ is true, i.e.,

$$F_{na} \equiv 0 \pmod{F_a}.$$

Base Case: $P(0) = F_0 \equiv 0 \pmod{F_a}$ Base case holds. We now prove $P(n+1)$, namely,

$$F_{(n+1)a} \equiv 0 \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \pmod{F_a}.$$

Using part (a) with n replaced by na ,

$$F_{na+a} \equiv F_{na} F_{a+1} \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \cdot F_{a+1} \pmod{F_a}.$$

$$F_{na+a} \equiv 0 \pmod{F_a}.$$

This proves $P(n+1)$ holds, and by the induction hypothesis $P(n)$ is true for $n \geq 0$. $\square$ Now if $a|b$, then $\exists k \in \mathbb{N}$ such that $a * k = b$. Now by $P(k)$, $F_{ak} = F_b \equiv 0 \pmod{F_a}$ which implies $F_a|F_b$.

Problem 2. Computing Modular Inverses

(a) First, using Euclid's algorithm:

$$\begin{aligned} \gcd(43, 15) &= \gcd(15, 13) & 13 &= 43 - 2 \cdot 15 \\ &= \gcd(13, 2) & 2 &= 15 - 1 \cdot 13 \\ &= \gcd(2, 1) & 1 &= 13 - 6 \cdot 2 \end{aligned}$$

Now doing backwards substitution (Pulverizer):

Substitute 2 (from $2 = 15 - 1 \cdot 13$)

$$\begin{aligned} 1 &= 13 - 6 \cdot (15 - 1 \cdot 13) \\ 1 &= 7 \cdot 13 - 6 \cdot 15 \end{aligned}$$

Substitute 13 (from $13 = 43 - 2 \cdot 15$)

$$\begin{aligned} 1 &= 7 \cdot (43 - 2 \cdot 15) - 6 \cdot 15 \\ 1 &= 7 \cdot 43 - 14 \cdot 15 - 6 \cdot 15 \\ 1 &= 7 \cdot 43 - 20 \cdot 15 \end{aligned}$$

The coefficient of 15 is -20. Therefore, the inverse would be $43 - 20 = 23$. So $23 \cdot 15 \equiv 1 \pmod{43}$.

(b) From Fermat's Little theorem we know that $15^{42} \equiv 1 \pmod{43}$ and now multiplying both side with 15^{-1} we get

$$15^{41} \equiv 15^{-1} \pmod{43}$$

Now to simplify 15^{41} we calculate the smaller powers of 15:

$$\begin{aligned} 15^1 &\equiv 15 \pmod{43} \\ 15^2 &= 225 \equiv 10 \pmod{43} \\ 15^4 &\equiv 100 \equiv 14 \pmod{43} \\ 15^8 &\equiv 196 \equiv 24 \pmod{43} \\ 15^{16} &\equiv 576 \equiv 17 \pmod{43} \\ 15^{32} &\equiv 289 \equiv 31 \pmod{43} \end{aligned}$$

Now $15^{41} = 15^{32} \cdot 15^8 \cdot 15$ Therefore, $15^{-1} \equiv 31 \cdot 24 \cdot 15 \equiv 23 \pmod{43}$ Meaning the inverse of 15 mod 43 is 23.

Problem 3. Primality Testing

(a)

Proof by Contradiction. Assume an integer $1 \leq a \leq n-1$ is fermat's witness of some prime number n . since a is fermat's witness $a^{n-1} \not\equiv 1 \pmod{n}$. Also since n is prime and a is in range $[1, n-1]$, Therefore a is not multiple of n so by fermat's little theorem $a^{n-1} \equiv 1 \pmod{n}$ which is a contradiction. $\square$

(b)

Proof. Assume p is a prime and $x^2 \equiv_p 1$ by the definition of congruence $p \mid x^2 - 1$

$$p \mid (x-1)(x+1)$$

Since p is prime (by Lemma 9.4.2) $p \mid (x-1)$ or $p \mid (x+1)$ so by definition of congruence $x \equiv_p -1$ or $x \equiv_p 1$. $\square$

(c)

Proof by contradiction. Assume, for the sake of contradiction, that the algorithm returns 'composite' for some prime number n . So at least one of the conditions from the algorithm must hold.

- $x_e \not\equiv_n 1$
- Consecutive terms (x_i, x_{i+1}) where $x_{i+1} \equiv_n 1$ but $x_i \not\equiv_n \pm 1$.

Suppose the first condition holds; then $x_e = a^{n-1} \not\equiv_n 1$. Now Since n is prime and $1 \leq a \leq n-1$ and Fermat's Little Theorem implies that $a^{n-1} \equiv_n 1$, which is a contradiction.

The first condition led to a contradiction, so the second condition must be true. So $x_{i+1} = x_i^2 \equiv_n 1$ but $x_i \not\equiv_n \pm 1$. Since n is prime, by lemma from part (b), the given condition yields a contradiction.

Both conditions lead to a contradiction, meaning our assumption must be wrong, so n must indeed be composite. $\square$

Problem 4. Pulverizer State Machine

(a) For some arbitrary state (x, y, s, t, u, v) say Pres1, Pres2, Pres3 are true and we need to prove that Pres1, Pres2, Pres3 are true for the next state (x', y', s', t', u', v') .

By the definition of transition in the given state machine, we have $x' = y$, $y' = (x \text{ rem } y)$, $s' = u$, $t' = v$, $u' = s - qu$, and $v' = t - qv$, where $x = qy + (x \text{ rem } y)$.

- **Pres1:** $\gcd(x', y') = \gcd(a, b)$ we know that $\gcd(x', y') = \gcd(y, x \text{ rem } y) = \gcd(x, y) = \gcd(a, b)$ which proves Pres1 is preserved.
- **Pres2:** $s'a + t'b = x'$, using the definition of s', t', x' we have $s'a + t'b = ua + vb = y = x'$ which proves Pres2 is preserved.
- **Pres3:** $u'a + v'b = y'$, using the definition of u', v', y' we have

$$\begin{aligned} u'a + v'b &= (s - qu)a + (t - qv)b \\ &= (sa + tb) - q(ua + vb) \\ &= x - q(y) \\ &= (x \text{ rem } y) \\ &= y' \end{aligned}$$

which proves Pres3 is preserved.

(b) From part (a), we proved that Pres1 and Pres2 are preserved, and from the definition of the given state machine we know that In the final state, $y = 0$, so now substituting y and Pres2 in Pres1, we get $\gcd(a, b) = \gcd(x, y) = \gcd(sa + tb, 0) = sa + tb$. Therefore, s and t are the coefficients satisfying Bézout's identity.

(c) Because the given state machine is just the Euclidean algorithm with some extra computation, and even the termination condition is same as the Euclidean algorithm, which is when $y = 0$.